A.L.T.R.U.S – Adaptive Life-support & Therapeutic Robotic Unit System

Project ID: 25-26j-437

Project Proposal

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B.Sc. (Hons) Degree in Information Technology

Department of Software Engineering

Sri Lanka Institute of Information Technology

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July 2025

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DECLARATION

We are confirming that the work given in this proposal is one of our own volition. The same work has not been given, entirely or partially, for academic credit at any other university or institution. We have also made sure that all sources of information and ideas from others have been clearly cited and appropriately acknowledged in the document.

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ABSTRACT

This article presents A.L.T.R.U.S (Adaptive Life-support & Therapeutic Robotic Unit System), a context-aware, modular assistive robot designed for safe, smart, and empathetic behavior in dynamic home scenarios. The system integrates four core software frameworks within a ROS2-based architecture to overcome the majority of shortcomings of current assistive robotics. An adaptive navigation module initially combines static SLAM with dynamic real-time LIDAR-based updates according to RSO2 logic to enable dynamic replanning of paths in response to environmental change—remapping is not needed. Voice and hand gesture control is integrated directly into the navigation layer so that users can call, pause, or drive the robot by natural voice commands or by a hand wave, enabling much enhanced accessibility for the disabled and elderly. Second, an environmental and health monitoring system in real-time combines data from environmental sensors (MQ-2, DHT11) and vital sign sensors (MPU6050, MAX30100) to detect emergencies like gas leaks, fever, or falls and provide automatic alerts via Firebase and Twilio. Third, an emotion-perceiving HRI module detects facial expressions and voice tone with OpenCV and pre-trained models (RAVDESS, FER2013), providing empathetic responses through a chatbot based on Dialogflow. Fourth, a fault-tolerant middleware offers system robustness through auto-recovery, intent arbitration, plugin update security, and blockchain log-based recording with Hyperledger Fabric for trust and transparency. Based on a modular ROS2 foundation, A.L.T.R.U.S brings together adaptive navigation, proactive healthcare, emotional intelligence, and robust software infrastructure in a unified, edge-deployable platform. This student innovation stretches the limits of assistive robotics by combining autonomy, safety, empathy, and security, which is ideal for smart homes, elderly care, and independent living support.

Keywords:

Adaptive Navigation, Dynamic SLAM, Human-Robot Interaction, Voice Command, Gesture Recognition, Emotion Recognition, Assistive Robotics, ROS2, Sensor Fusion, Blockchain, Proactive Healthcare

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LIST OF ABBREVIATIONS

ABBREVIATION	DESCRIPTION
SLAM	Simultaneous Localization and Mapping
ROS2	Robot Operating System 2
LIDAR	Light Detection and Ranging
NLP	Natural Language Processing
HRI	Human-Robot Interaction
STT	Speech-to-test
GUI	Graphical User Interface
Gazebo	Robot Simulation Environment

1. INTRODUCTION

1.1 BACKGROUND

The home care future lies in smart, empathetic, and learning-enabled robots—platforms that not only support and move but also understand, respond, and learn with users. To meet the growing demand for affordable, personalized, and decentralized care, this project proposes a next-generation assistive robot for deployment within dynamic home environments.

Drawing inspiration from emotionally intelligent robots such as Baymax, our framework combines four fundamental software building blocks in one coherent, ROS2-based system:

Voice-Based Command & Adaptive Navigation:
This module combines RPLIDAR-based SLAM with dynamic replanning of paths using reinforcement learning to enable safe navigation around evolving home settings. The robot re-routes in real time when objects are encountered or furniture is shifted by responding to voice instructions, enabling non-stop autonomy even in dynamic settings.

Real-Time Health & Environmental Monitoring
The framework combines sensor data from wearables and the environment (e.g., heart rate, temperature, gas concentrations) to monitor user well-being in real time. Upon emergencies such as falls, fever, or leaks of gas, the framework triggers automatically generated alerts via cloud services like Firebase and Twilio, enabling remote caregivers to respond immediately. This feature offers proactive monitoring and timely intervention.

Emotion Detection & Empathetic Behavior Generation:
With the implementation of technologies such as OpenCV and TensorFlow, the system reads facial expressions and voice tone to identify emotions like sadness, worry, or suffering. A chatbot interface generates empathetic responses (e.g., "You look upset. Do you want to talk?"), with users empowered to converse nonverbally through gesture recognition. The multimodal approach enhances natural interaction and emotional response, and the robot is that much more intuitive and useful for technical or elderly users.

Fault-Tolerant Middleware with Secure Plugin Updates:
Built on ROS2, the system possesses a fault-tolerant, modular middleware that supervises all components. It provides support for intent-based behavior arbitration (resolution of conflicting commands) and secure, blockchain-logged plugin updates, ensuring reliability and trust for mission-critical healthcare functions. This architecture enables strong runtime recovery, dynamic skill composition, and secure communication, making the system strong and scalable to evolving needs.

This merged system addresses key limitations in current assistive robots, including inflexible navigation, mechanical interaction, health insensitivity, and vulnerability to breakdowns.

Through incorporating adaptive navigation, natural human-robot interaction, real-time monitoring of health, and fault-tolerant middleware, the resulting system provides context awareness, emotional intelligence, and self-sustenance for practical deployment. It transforms traditional assistive robots into intelligent companions that can dynamically adapt to changing environments while offering natural, multimodal control interfaces—all within a modular, scalable framework like ROS2 for real-world assistive applications.

The Rise of Assistive Robotics in Smart Homes

The integration of robotics into domestic environments has accelerated in recent years, driven by advancements in artificial intelligence (AI), sensor technology, and edge computing. Home assistive robots are now increasingly used to help elderly individuals, people with disabilities, and busy households perform tasks such as object delivery, monitoring, and providing companionship.

A fundamental requirement for these robots is the ability to navigate safely and autonomously within complex and often unpredictable indoor spaces. Many commercially available systems use Static Simultaneous Localization and Mapping (SLAM), which creates a fixed map during an initial setup phase and uses it for future navigation. While effective in stable environments, this approach struggles in real homes where changes frequently occur — such as furniture rearrangement, closed doors, or the presence of pets and toys. In these cases, static maps become outdated, leading to navigation errors, manual interventions, and reduced trust in the system.

To address these challenges, researchers have developed dynamic path planning and real-time obstacle detection techniques. While these approaches improve adaptability, many existing solutions are limited in scope, lack full integration with user-friendly interaction systems, or are not specifically designed for assistive purposes.

An equally important aspect of assistive robotics is Human-Robot Interaction (HRI). Elderly and non-technical users often find traditional control methods, such as mobile applications or physical remotes, challenging to use. Natural interaction methods — such as voice commands and gesture recognition — offer more intuitive and accessible ways to control robots. Technologies like Vosk and DeepSpeech (for speech recognition) and OpenCV with MediaPipe (for gesture detection) have made such interfaces more feasible in real-time applications.

Despite progress in both navigation and interaction technologies, there remains a clear need for a unified system that can dynamically adapt to environmental changes while offering intuitive, multimodal control methods — all implemented in a modular, scalable framework like ROS2 for real-world assistive use.

User Feedback to Support the Research

To validate the practical need for such a system, a user survey was conducted with 20+ participants.

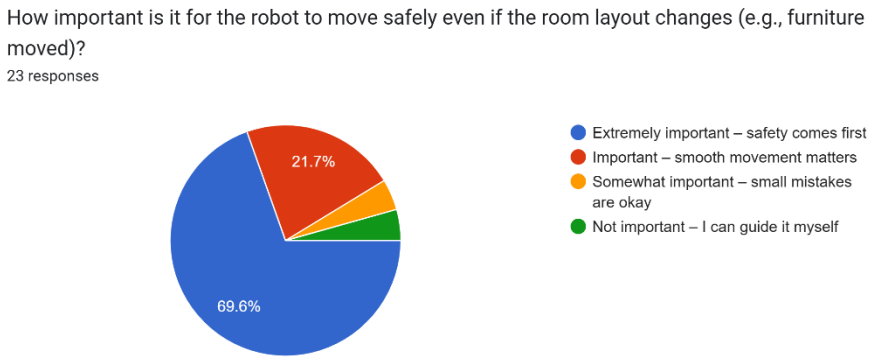


Figure 1.1.1 – Survey results on importance of safe movement in changing room layouts

Figure 1.1.1 illustrates survey findings on the importance of safe movement in dynamic home environments. All participants (100%) indicated that it is very important for a robot to navigate safely even if the room layout changes, such as when furniture is moved or obstacles appear unexpectedly. This highlights the necessity of adaptive navigation systems that can operate reliably without manual intervention.

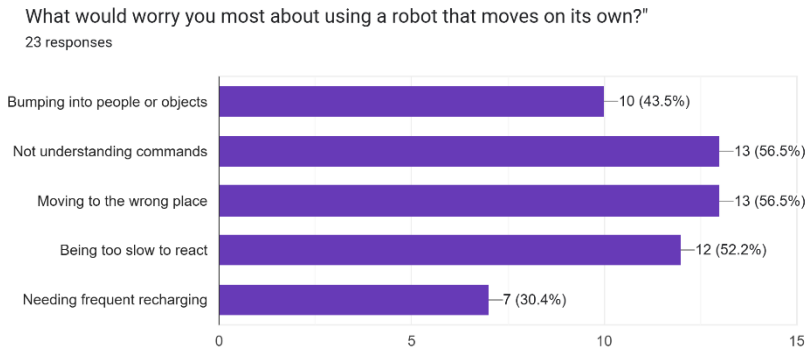


Figure 2.1.2 – Survey results on main concerns about autonomous robots

Figure 2.1.2 summarizes participants' main concerns about using an autonomous robot. The top concern, reported by all respondents (100%), was bumping into people or objects. Other concerns included not understanding commands (82%), moving to the wrong place (68%), being too slow to react (45%), and needing frequent recharging (36%). These responses underscore the importance of collision avoidance, accurate command interpretation, and responsive decision-making in assistive robots.

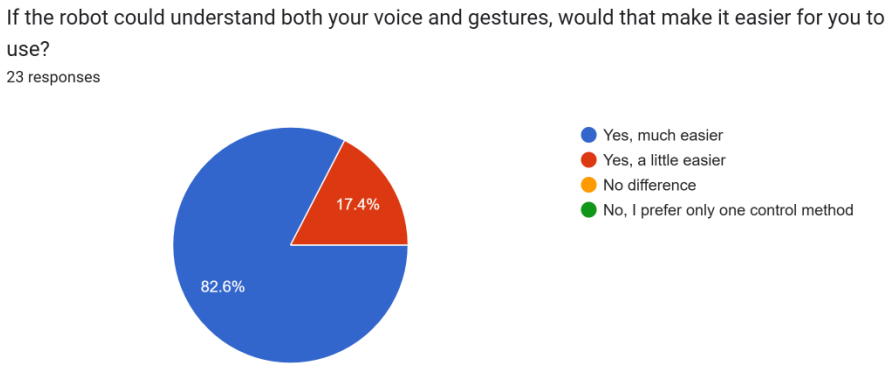


Figure 3.1.3 – Survey results on preference for voice + gesture control

Figure 3.1.3 shows the level of interest in multimodal control methods. A large majority of participants (91%) indicated that the ability for the robot to understand both voice and gesture commands would make it easier to use. This confirms the strong user demand for intuitive, natural interaction methods, especially in contexts involving elderly or physically challenged users.

These findings confirm that there is a strong user demand for intelligent, responsive, and easy-to-use assistive robots that can adapt to changing environments and provide accessible multimodal control.

In conclusion, while individual technologies for navigation and interaction exist, there remains a need for a comprehensive, integrated system that combines adaptive navigation, real-time obstacle handling, and natural multimodal interaction — all implemented in a robust, modular framework like ROS2, without relying on cloud processing or complex user interfaces.

1.2 LITERATURE SURVEY

This section provides an overview of the latest research and technology in terms of adaptive navigation and human-robot interaction for assistive robotics and their strengths and limitations compared to the system discussed here.

1.2.1 Navigation Systems and SLAM Approaches

Static SLAM – Studies such as the TurtleBot3 navigation system [1] and Zhang et al. (2021) [2] utilized GMapping-based static SLAM efficiently in indoor mapping and localization. Even though such systems are efficient in stationary environments, they do not adapt when there is a change in setup, hence requiring remapping manually and limiting real-world application in dynamic home settings.

Dynamic SLAM – The ROS2 Nav2 navigation stack [3] is an example of systems that utilize additional sensors such as LIDAR or depth cameras to dynamically detect obstacles and navigate around them. These systems allow local costmap update and real-time path replanning, but usually operate independently without high-level interaction modules, so users cannot control navigation by natural means such as voice or gesture.

Hybrid Approaches – A few studies, such as that of Sitaraman and Khalid (2024) [4], present hybrid navigation systems that combine static SLAM with global localization and dynamic sensing for real-time adaptation. Their approach combines AutoNav, LIDAR-based SLAM, and a DenseNet-based motion planner with Leaky ReLU activation, achieving 95% gradient efficiency, 0.045 m localization error, and 97.5% pathfinding accuracy. While these results show great technical performance, the system is primarily geared towards industrial and research environments and is not suited for addressing the demands of assistive home robotics or integrating multimodal interaction methods such as voice and gesture control.

1.2.2 Human-Robot Interaction (HRI)

Voice Interfaces – Speech recognition toolkits such as Vosk and DeepSpeech have been applied in assistive robotics to enable voice-controllable access. For instance, Fuad et al. (2024) presented an open-source, offline voice command–based system built with the assistance of Vosk, DeepSpeech, and CMUSphinx. The system achieved a 91% success rate in real-time operation, being robust and real-world viable—especially in supporting users with mobility disabilities [5].
While these systems expand access through voice, they must contend with phenomena like background noise and room acoustics. Proper deployment in home settings therefore requires robust noise management and greater contextual sensitivity to ensure accurate and reliable performance. [6]

Gesture Recognition – A study by Kim et al. (2022) and Patel et al. (2023) illustrates how libraries of computer vision like OpenCV and MediaPipe can enable robots to interpret hand signs for gesture-based touch-free control. However, the majority of such systems are restricted to pre-determined gestures and do not integrate gesture recognition with adaptive navigation logic.

Multimodal Interfaces – Scholars such as Ferrari et al. (2023) [7] and Vanc and Stepanova (2025) [8] investigated multimodal fusion of voice and gestures in human–robot cooperation. Ferrari et al. [7] present a multimodal fusion system that works through voice control and gesture recognition to add flexibility and safety to collaborative robots. Similarly, Vanc and Stepanova [8] introduce TransforMerger, a transformer model with the capability of fusing voice and gesture information into a single, structured command with greater noise robustness and uncertainty in input. While these approaches are state-of-the-art polyphonic multimodal interaction, they still treat voice and gesture as separately controlled streams with no full decision and conflict-resolution process in integration. As a result, the systems can engage in undesired or inconsistent behavior for intricate situations-especially for dynamic, home assistive settings where navigation and interaction need to be highly coupled.

1.2.3 Shortcomings of Existing Literature

On the basis of the literature analyzed, the following shortcomings are evident:

- The majority of domestic robots employ fixed SLAM with no real-time adaptation to dynamic environments.
- Dynamic navigation supporting systems will not usually offer intuitive, natural interaction interfaces.
- Voice and gesture systems are rarely blended together with adaptive navigation in a single architecture.

Few solutions are rooted in modular, scalable structures like ROS2, which allow the potential for future expansion and interoperability.

The system presented fills this gap by designing a ROS2-based hybrid navigation framework that integrates static + dynamic SLAM, real-time LIDAR-based obstacle management, and multimodal interaction (voice + gesture) - particularly for assistive home environments.

1.3 RESEARCH GAP

In spite of dramatic improvements in robotics, SLAM, and HRI, current robotic systems are still far from achieving robustly autonomous, adaptive, and user-friendly performance in dynamic home environments—particularly for elderly or physically impaired users.

Most robots today are designed for static, pre-mapped environments and rely heavily on fixed paths of navigation. When the environment changes—furniture is moved, doors are shut, or toys are on the floor—these robots fail to replan their path and therefore navigate, cause user frustration, or require manual correction. Further, interaction methods remain limited to mobile apps, remote controls, or mere voice commands, not intuitive for non-technical users.

To identify the research gap, we compared six model systems in more recent works to our solution. The results are shown in Table 1.3.1.

Feature	Research [1]	Research [2]	Research [3]	Research [4]	Research [5]	Research [7]	Proposed System
Static SLAM Only	**Yes**	**Yes**	**No**	**No**	**No**	**No**	**No**
Dynamic Path Update	**No**	**No**	**Yes**	**No**	**No**	**No**	**Yes**
Hybrid SLAM (Static + Dynamic)	**No**	**No**	**No**	**Yes**	**No**	**No**	**Yes**
Voice Command Support	**No**	**No**	**No**	**No**	**Yes**	**Yes**	**Yes**
Gesture Recognition	**No**	**No**	**No**	**No**	**No**	**Yes**	**Yes**
Multimodal Fusion (Voice + Gesture)	**No**	**No**	**No**	**No**	**No**	**No**	**Yes**
Assistive Focus (Elderly/Home Use)	**No**	**No**	**No**	**No**	**Yes**	**No**	**Yes**
ROS2-Based Modular Design	**Yes**	**No**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

Table 1.3.1 – Comparative analysis highlighting the novelty of the proposed system

[1] TurtleBot3 Navigation using Static SLAM

This paper presents a fundamental implementation of GMapping-based SLAM on the TurtleBot3 robot system, where the robot can build a 2D occupancy grid map of the environment and find itself in the map. It relies on odometry and LIDAR data to perform Simultaneous Localization and Mapping (SLAM) in indoor spaces, demonstrating robust performance in stationary, obstacle-free scenes. But the system relies on the assumption that the environment is invariant after the process of initial mapping. It has no dynamic obstacle sensing and online path replanning. New obstacles (e.g., a chair moved into the path of travel) are either not seen or result in robot stalling, requiring human intervention. It also lacks human-robot interaction, hence making it inaccessible to non-expert users such as the disabled or elderly. This is a significant limitation: while the system is perfect for education and straightforward navigation use, it is not suited to domestic home settings, which are dynamic in nature. Our proposed system overcomes this by including real-time LIDAR-based obstacle detection and dynamic path updates, ensuring constant autonomy.

[2] PLD-SLAM: RGB-D SLAM for Dynamic Indoor Environments

This paper introduces PLD-SLAM, a visual-inertial SLAM algorithm that uses RGB-D cameras to recover point and line features for robust mapping of indoor environments. It concentrates particularly on dynamic scenes via the detection and rejection of moving objects (human beings, for instance) in an attempt to maintain a clean static map for localization. Though this is a crucial step toward controlling dynamic behavior, the system is not designed for navigability or real-time path modification. Its primary focus is map accuracy, and not robot mobility or user control. It does not generate collision-free paths as novel static obstructions (e.g., furniture) arise, nor voice nor gesture commands. Also, it relies on good camera input and can break down in bad lighting. Our design leans upon this utilizing 2D LIDAR (stronger in fluctuating light) and combining static mapping with dynamic costmap updates to facilitate adaptive navigation—not just robust mapping.

[3] ROS2 Nav2 Navigation System ("The Marathon 2")

This piece of writing presents Nav2, ROS2's second-generation navigation stack that is highly utilized in service and industrial robots. Dynamic obstacle avoidance, local costmap revision, and real-time replanning with LIDAR and other sensors are enabled by Nav2. It is highly modular, thus developers can readily plug in recovery behavior, custom planners, and controllers. As technically advanced as it is, Nav2 is not designed to be used directly by end-users. It assumes that navigation goals will be provided programmatically or via a user interface instead of naturally by humans. There is no built-in support for voice or gesture input, and the system does not attempt to optimize for accessibility by the elderly or physically disabled. Our proposed system is an enhancement of Nav2 but with the inclusion of a multimodal interaction layer, so users can now simply command or wave to activate follow mode. This turns Nav2 into a user-centered assistive system from a developer-centered tool.

[4] Adaptive Motion Planning Hybrid SLAM

The paper introduces a hybrid navigation system by integrating static SLAM with deep learning-based motion planning using DenseNet and Leaky ReLU. The system employs LIDAR data to detect obstacles and uses a neural network to predict optimal motion trajectories, improving adaptability in dense scenes. Deep learning for path planning is new but without any integration of high-level human commands. The neural network is a black-box controller, difficult to interpret and amend according to user commands. It also lacks voice or gesture recognition and lacks any reference to assistive or healthcare-based applications. Our system is more modular and explainable, using rule-based behavior trees and decision engines to integrate sensor data with natural user input, giving safe navigation and explainability in assistive environments.

[5] Open-Source Voice Command System for HRI

This project presents a lightweight, offline voice command system based on Vosk and DeepSpeech for converting speech to text. It achieves real-time accuracy of 91% and can be used for resource-constrained devices and hence ideal for embedded robotics. However, the system has no pre-defined commands beyond and isn't even coupled with navigation logic. For example, giving the command "Go to kitchen" does not make a navigation occur—it only logs the command. There is no connection to a path planner or SLAM system, and gesture input is not present. Our system addresses this constraint by directly mapping voice commands to navigation targets. We also integrate voice with gesture input to allow users to gesture to a destination or wave to trail, offering an entirely multimodal interface.

[7] Safe Multimodal Communication in Human-Robot Collaboration

This research explores multimodal HRI by fusing voice and gesture inputs to improve safety and flexibility in collaborative robotics. While it is multimodal input supportive, it does not arbitrate voice and gesture command conflicts. It also lacks integration with adaptive navigation in home settings.

1.4 RESEARCH PROBLEM

Most domestic robots nowadays employ static SLAM (Simultaneous Localization and Mapping), generating a static map of the space during an initial setup and using it for later navigation. While such a strategy performs well in stable and predictable environments, it performs poorly in dynamic home environments where changes are frequent - e.g., furniture is moved, doors are open or closed, or toys and other temporary obstacles are present on the floor. Static maps then become outdated, preventing the robot from recalculating optimal paths in real time. This leads to navigation errors, collisions, task failure, and a decrease in the level of user trust.

Apart from navigation, interaction constraints also reduce the usability of assistive robots. Most systems solely employ mobile apps or physical remote controls, which can be non-intuitive for elderly people, individuals with disabilities, or non-technical users. Although there are robots that include voice or gesture recognition, these are typically included as independent features without integration into the navigation logic and thus cannot be utilized for adaptive, context-aware decision-making.

Some of the remaining challenges in assistive robotics include:

- Real-time flexibility — sensing and compensating for environmental changes without full remapping.
- Multimodal command fusion — fusing voice and gesture inputs into a unified control system with conflict resolution.
- Operational safety — ensuring reliable navigation in cluttered homes with children, pets, and shifting layouts.
- Accessibility — providing natural and intuitive modes of interaction that do not require a high degree of technical expertise.

It is the intention of this research to address these limitations through the development of an ROS2-based hybrid navigation system that fuses static SLAM-based global positioning with dynamic LIDAR-based obstacle detection for local adaptation in real time. Multimodal interaction shall be incorporated into the system, enabling voice and gesture commands to naturally complement each other, making the robot more secure, autonomous, and convenient to operate in practical home environments.

2. RESEARCH OBJECTIVES

2.1 MAIN OBJECTIVE

The main objective of this work is the design, implementation, and evaluation of an adaptive navigation and multimodal human–robot interaction system for assistive home robots. The system will enable full autonomous operation in dynamic, unstructured indoor environments by combining hybrid SLAM navigation with natural, intuitive control methods—specifically voice and gesture commands. This approach aims to significantly enhance accessibility, safety, and operational efficiency, making the technology more usable by elderly consumers, physically impaired individuals, and non-technical users.

2.2 SPECIFIC OBJECTIVES

Implement a Hybrid SLAM Navigation System

- Merge static SLAM for accurate global localization with real-time LIDAR-based obstacle detection for local adaptation.
- Enable dynamic path replanning for the robot to respond to alterations in the layout-such as moved furniture or unexpected obstacles—without requiring full remapping.

Develop a Voice Command Processing Module

- Use speech-to-text (STT) technology for understanding natural voice commands.
- Use context-aware command interpretation for handling variation in wording, reducing confusion, and ensuring safe execution in the presence of noise.

Build a Gesture Recognition System

- Use computer vision with OpenCV and MediaPipe for detecting predefined gestures.
- Engineer the system to function well under varying light conditions and camera angles for steady recognition.

Engineer a Central Command Handler

- Create a decision engine that integrates both voice and gesture module inputs into a single control system.
- Incorporate conflict resolution mechanisms to allow the system to prioritize commands and prevent unsafe actions.

Enable Autonomous Docking

- Develop a docking routine to allow the robot to locate and drive to its charging station using landmark recognition and precise alignment algorithms.
- Ensure the docking operation is robust under different lighting and floor surface conditions.

Monitor and Respond to Changes in the Environment

- Continuously process LIDAR and vision sensor information to detect significant changes in the environment.
- Trigger incremental replanning or partial remapping processes without interrupting ongoing tasks.

Ensure System Reliability, Safety, and Usability

- Conduct thorough testing in both simulated environments (e.g., ROS2's Gazebo) and real home environments.
- Measure performance in terms of navigation accuracy, obstacle avoidance success rate, docking success rate, command recognition accuracy, and user satisfaction ratings.

3. METHODOLOGY

The proposed system enhances home assistive robots' autonomy and functionality with the implementation of Hybrid SLAM navigation, real-time obstacle avoidance, and multimodal human–robot interaction (HRI) through voice and gestures. The design is accomplished entirely in a ROS2 modular architecture for scalability, maintainability, and fault tolerance.

OVERALL SYSTEM DIAGRAM

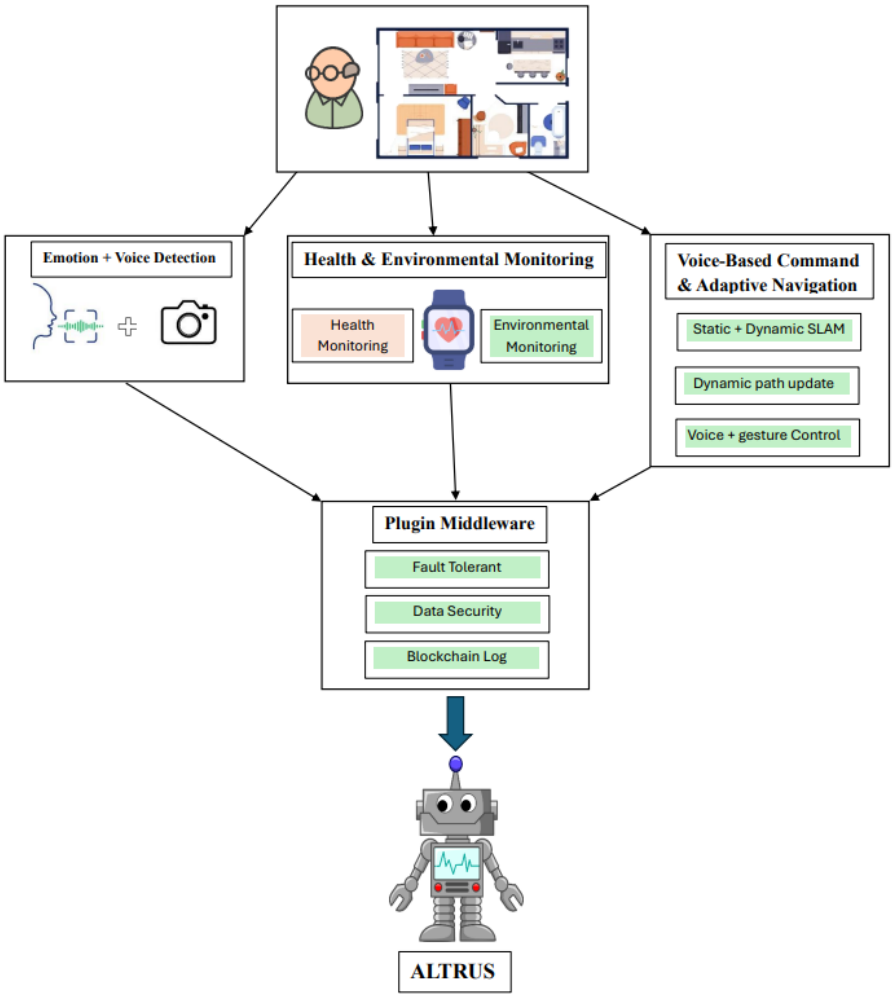


Figure 4.3.1 System Overview Diagram

MY COMPONENT SYSTEM DIAGRAM

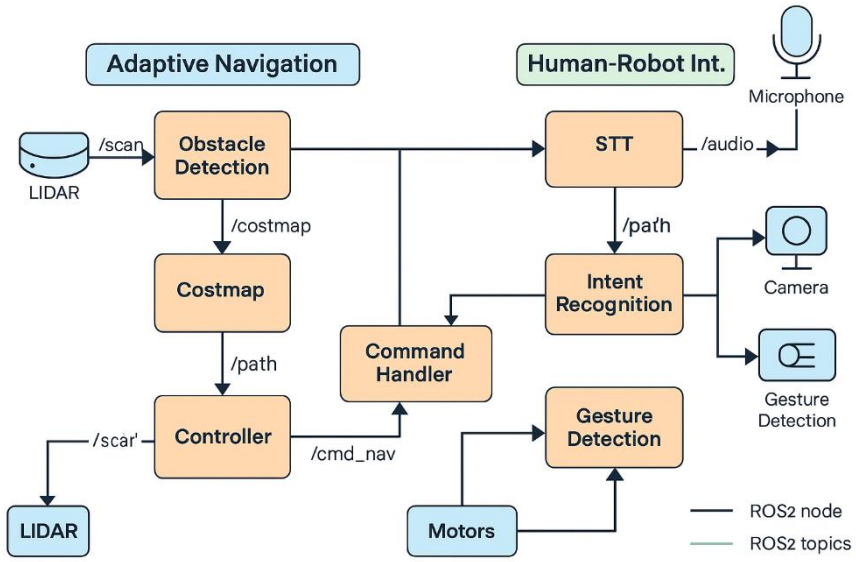


Figure 5.3.2 Component system diagram

The system diagram illustrates the architecture of the suggested Adaptive Navigation and Human–Robot Interaction (HRI) system for assistive home robots, designed as a modular ROS2 system. The organization includes three main functional groups:

Adaptive Navigation Module – One LIDAR sensor provides real-time scan of the robot's environment and feeds the Obstacle Detection node with real-time information. The observed changes from the static SLAM map are forwarded to the Costmap node, updating the robot's environment model. The updated path is utilized by the Controller to produce motion commands for collision-free navigation, and the motors execute the movement.

Human–Robot Interaction Module – Voice is received by a microphone and processed within a Speech-to-Text (STT) node to convert what is being said into text. The text is then processed within the Intent Recognition node to identify navigation targets. A camera takes simultaneous visual information for the Gesture Detection node, which detects certain hand gestures.

Command Handling Layer – The Command Handler serves as the point of decision-making, receiving input from the intent recognition (speech) and gesture detection nodes. It prioritizes and reconciles conflicting commands before issuing final instructions to the navigation controller.

Through sensor-based navigation coupled with multimodal user interaction within an ROS2-based environment, the system enables autonomous, safe, and user-friendly operation in dynamic and unpredictable home environments.

3.1 Environment Navigation and Mapping

Initial Localization

The robot starts by downloading a pre-computed static map of the home environment, which is acquired through static SLAM during an initial mapping procedure.

Global localization is performed to define the robot's starting point in map terms.

Real-Time Obstacle Detection and Adaptation

A 2D LIDAR sensor is constantly observing the environment to search for deviations from the static map, e.g., new obstructions (e.g., furniture, items on the floor) or dynamic environments.

Detected changes are merged into a local costmap.

Dynamic Path Replanning

If it detects an obstacle, the system updates the costmap and invokes a dynamic replanning algorithm (via ROS2 Nav2).

A new collision-free path is calculated without fallback to a full remapping, offering smooth, continuous navigation.

3.2 Human–Robot Interaction (HRI)

Voice Command Interface

User voice instructions are picked up by a microphone array.

Speech is processed offline by Speech-to-Text (STT) models such as Vosk to ensure privacy and offline functioning.

The text is converted to pre-defined navigation intents.

Gesture Recognition Interface

A camera module streams video input continuously.

OpenCV and MediaPipe frameworks process the input to identify gestures

Recognizable gestures are converted to equivalent action commands.

3.3 Central Command Handler

Both voice and gesture inputs are fed into a central decision engine.

This handler accomplishes

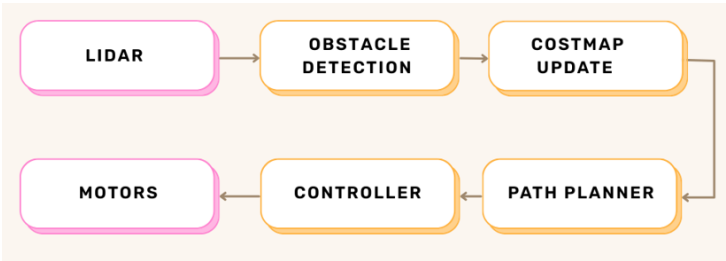
Conflict resolution.

Priority handling to deal with simultaneous inputs.

Command dispatch to the navigation system.

3.4 System Architecture Overview

Navigation Flow:



Voice + Gesture Flow:

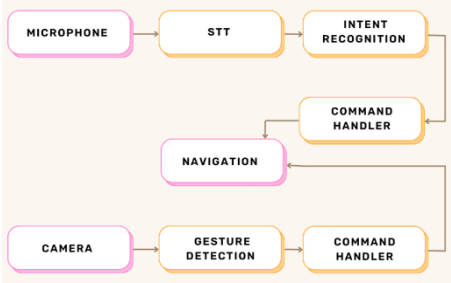


Figure 6.3.3 Adaptive Navigation and HRI Pipeline

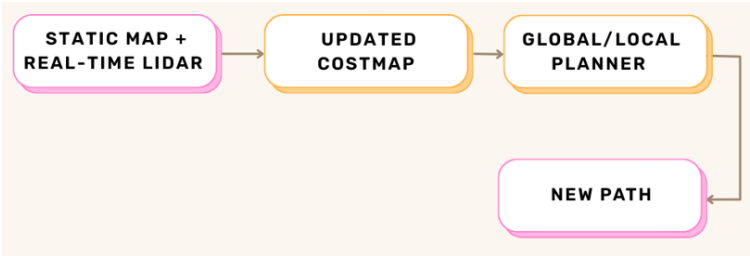


Figure 7.3.4 Hybrid SLAM and Dynamic Path Update Mechanism

3.5 Project Flow

3.5.1 Requirement Gathering and Analysis

The requirement phase consisted of meetings with supervisors, literature review, and an online survey of prospective users (N=20+). The survey assessed preferences for robot control methods, dynamic navigation importance, and interest in using voice/gesture interfaces. Results confirmed high demand for adaptive and intuitive robots.

Current systems were examined for gaps, prompting the prioritization of hybrid SLAM, real-time obstacle detection, and multimodal interaction.

3.5.2 Feasibility Study

Scheduled Feasibility: A Gantt chart was developed to control the 6-month schedule, encompassing design, implementation, and testing.

Technical Feasibility: The project employs mature technologies (ROS2, Vosk, OpenCV, Nav2) with accessible documentation and community support.

Economic Feasibility: The system is aimed at low-end hardware (Raspberry Pi, Jetson Nano, RPLIDAR). Development and deployment expenses are minimized.3.2.2 Feasibility Study

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3.5.3 Implementation

The system is implemented in Python and C++ with ROS2 (Humble).

Key components include:

SLAM and Navigation Stack (Nav2)

Voice Command Node (Vosk)

Gesture Recognition Node (OpenCV + MediaPipe)

Command Handler (Behavior Tree Logic)

Environment Monitor (LIDAR Data Analyzer)

All nodes communicate using ROS2 topics and services.

3.5.4 Testing

Testing includes:

Unit Testing : Each node is tested independently.

Integration Testing : Entire system tested on Gazebo simulation.

Real-World Testing : Deployed on a real robot in a dynamic home-like environment.

User Testing : Usability feedback from elderly volunteer users.

3.6 Commercialization

A.L.T.R.U.S (Adaptive Life-support & Therapeutic Robotic Unit System) will be promoted not only as an end-to-end integrated robot system but also as an open, modular set of software frameworks aimed at enabling developers and fostering innovation in assistive robots. Targeted markets are care homes for the elderly, smart homes, hospitals, rehabilitation facilities, and notably, robot start-ups and research institutions seeking solid, production-ready software building blocks.

In its core, A.L.T.R.U.S offers four independent but integrated ROS2-based software frameworks for plug-and-play deployment:

Navigation Framework : Active SLAM and light-weight reinforcement learning-based dynamic path planning within hybrid SLAM.

Health & Environmental Monitoring Framework : Predictive emergency detection using real-time sensor fusion (falls, fever, gas leak).

Emotion & Voice Interaction Framework : Multi-modal emotion recognition (face + voice + gesture) with empathetic response generation.

Fault-Tolerant Middleware Platform : Secure, self-healing core with intent arbitration, plugin control, and blockchain-based logging.

They will be shipped as developer kits (SDKs) with complete documentation, APIs, and model simulation-ready so that robotic startups and R&D organizations can easily incorporate advanced features into their own systems—without having to recreate from scratch. This "robotics enabler" strategy lowers the barriers to startup and accelerates time-to-market.

Commercialization will target:

- Direct sales of full A.L.T.R.U.S robot to care homes and smart houses.
- Upgrade software packages in modules (e.g., "Empathy Pack", "Health Sentinel Module") for incorporation into current service robots.
- Licensing and partnerships with robotic startups and universities for software framework adoption.
- Pilot tests in real care environments for safety, reliability, and user acceptability.
- Visibility through robotics conferences (e.g., ICRA, IROS), technology fairs, and university partnerships.

By placing A.L.T.R.U.S both as a physical care robot and as a robust software ecosystem, this project closes the gap between industrial and academic research—aligning the future of intelligent, empathetic, and safe assistive robots with reality.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

- The system must load a startup static SLAM map for initial localization.
- The system must detect new obstacles using LIDAR and update navigation paths in real time.
- The system must detect voice commands and execute corresponding navigation tasks.
- The system must detect hand gestures and respond accordingly.
- The system must resolve gestural vs. voice input conflicts.
- The system must dock automatically to charging station on battery low.
- The system must monitor environment change and replan if necessary.

4.2 Non-functional Requirements

- **Performance**: Must respond to commands in less than 2 seconds; updates to navigation in less than 500ms.
- **Reliability**: Must operate continuously for 8 hours; recover from sensor failures gracefully.
- **Security**: No personal data is stored; all is encrypted.
- **Usability**: No technical knowledge required; voice/gesture control intuitive.
- **Maintainability**: Good documentation and version control (Git) with modular design.

4.3 User Requirements

- User needs to have access to a tablet or smartphone for installation.
- User can be relied upon to communicate in simple English instructions.
- User can be expected to perform simple hand gestures.
- Robot must operate in a Wi-Fi-enabled environment.

5. BUDGET DETAILS

Item	Estimated Cost (LKR)
Raspberry Pi 5 8GB Original UK	Rs.39,950.00
RPLiDAR 2D	Rs.37,900.00
PK-910H 1080p Full-HD WebCam	Rs.8,000.00
Wheels	Rs.5,000.00
Server motor 2 (Mg90s)	Rs.1,600.00
Gear motor 2 (L298)	Rs.6,000.00
Motor controller	Rs.500.00
ESB 32	Rs.1,700.00
Oled display	Rs.600.00
Heart rate + blood oxygen	Rs.600.00
Ecg	Rs.1,600.00
Accelerometer + gyroscope	Rs.1,500.00
body temperature	Rs.2,500.00
lipo battery	Rs.2,500.00
Gas sensor + Fire sensor	Rs.1,000.00
Additional Cost	Rs.30,000
Total Cost	Rs.140,950.00

Table 2.1 Budget for the proposed system

6. CONCLUSION

The A.L.T.R.U.S is a significant step forward in the design of safe, intelligent, and empathetic assistive robots for real home healthcare environments. By integrating four leading-edge software paradigms—voice-based intent-based navigation, real-time environmental and health sensing, multi-modal emotion and gesture detection, and a fault-tolerant blockchain-encrypted middleware—this system abolishes the crippling weaknesses of assistive robots today, i.e., being limited to stationary configurations, with artificial interaction, and collapsing under dynamic or emergency situations. A.L.T.R.U.S offers a hybrid SLAM approach mixing static mapping with real-time LIDAR-based path adjustment through RSO2 logic, enabling the robot to re-route dynamically because of furniture re-arrangement or other obstructions—without complete remapping. This is further augmented with voice command control of navigation and gesture control, making it possible for elderly or physically disabled users to control naturally and intuitively, with just voice or gestures. In parallel, the system continuously monitors principal vital signs (heart rate, SpO_2, activity) and environmental parameters (temperature, humidity, gas level) with a ROS2-based sensor fusion framework and automatically triggers alarms through Firebase and Twilio upon detecting emergencies such as falls, fever, or gas leaks. The emotion-aware HRI module integrates facial expressions, voice tone, and body postures with OpenCV, TensorFlow, and pre-trained models to generate empathetic responses via a chatbot-based interface, improving interaction human-likeness and warmth. Essentially, the secure and self-healing middleware ensures system reliability through fault auto-recovery, intention arbitration of conflicting requests, safe plugin upgrades, and tamper-proof logging using Hyperledger Fabric blockchain—ensuring trust, transparency, and long-term evolvability. Fully developed on a modular ROS2 framework, A.L.T.R.U.S is not just scalable and extendable but also a student-developed innovation that bridges the gap between research at universities and real-world, practical application. This harmonious integration of adaptive mobility, active health monitoring, emotional intelligence, and robust software infrastructure makes A.L.T.R.U.S a pioneering platform in assistive robotics—a harbinger of a time when intelligent robots will finally provide independent, dignified, and connected living.

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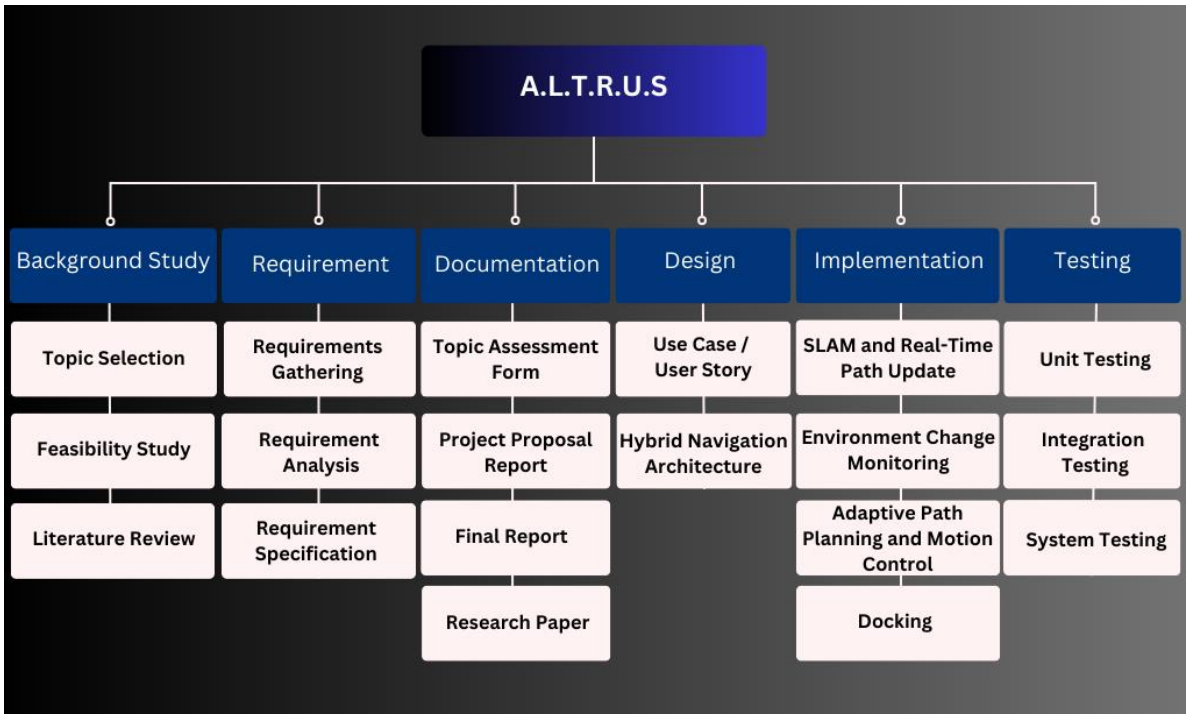
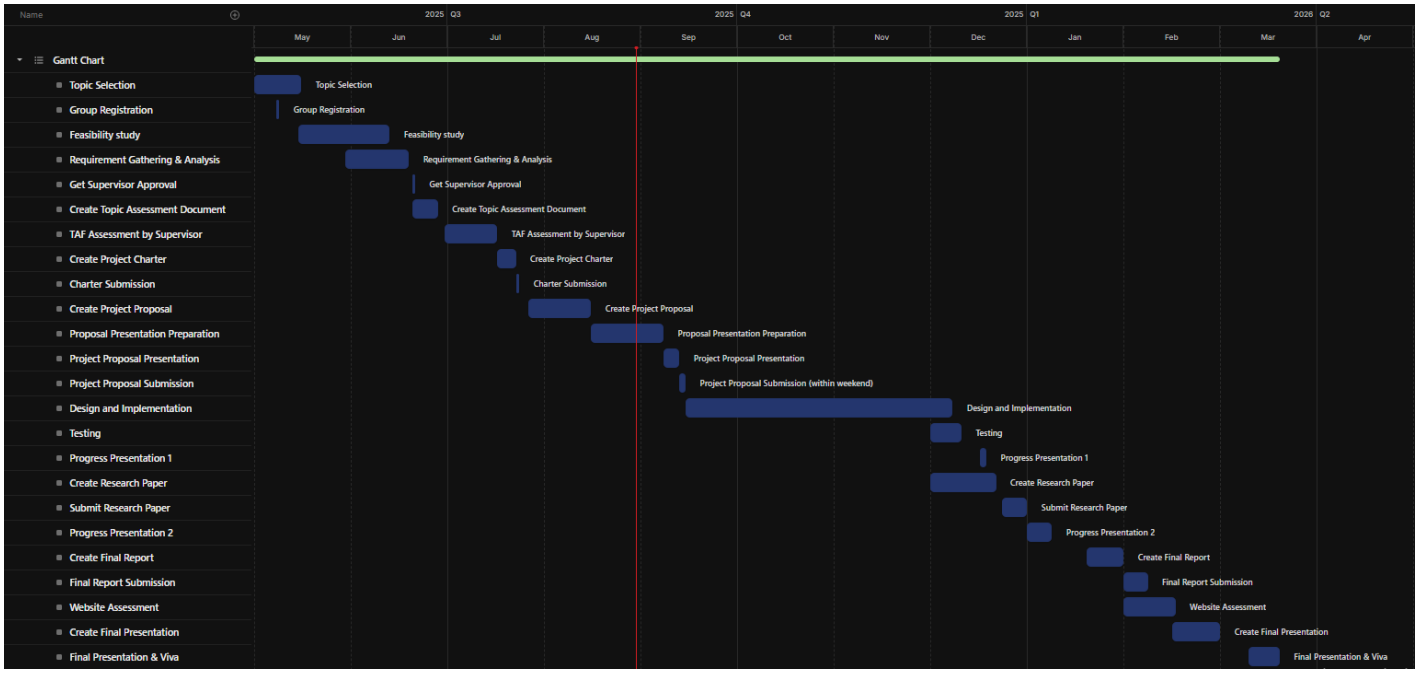
APPENDICES

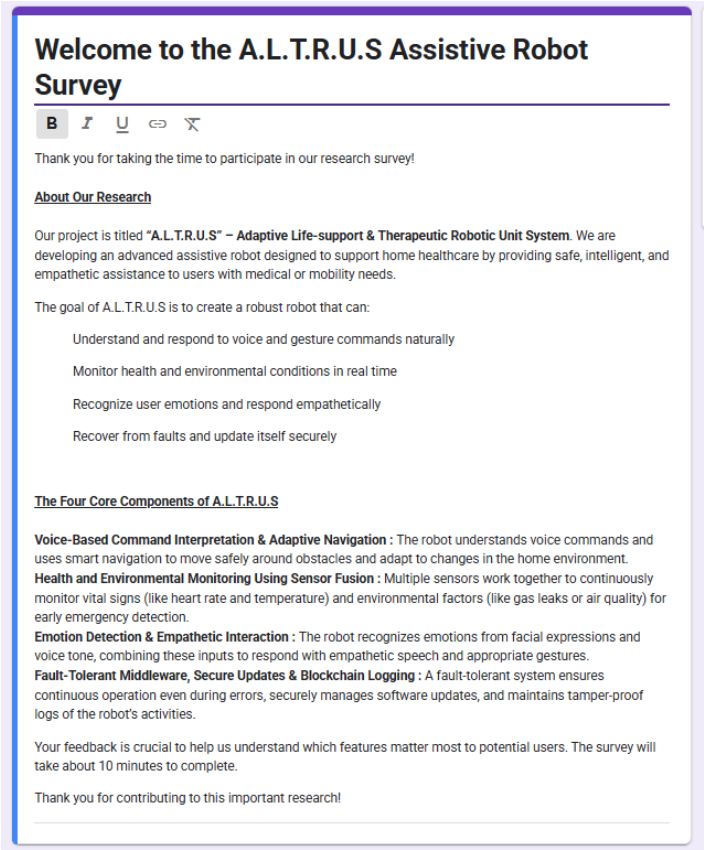
Appendix 1. Work Breakdown Structure (WBS)



Appendix 2. Gantt Chart



Appendix 3. Online Survey



Welcome to the A.L.T.R.U.S Assistive Robot Survey

Thank you for taking the time to participate in our research survey!

About Our Research

Our project is titled **"A.L.T.R.U.S" – Adaptive Life-support & Therapeutic Robotic Unit System**. We are developing an advanced assistive robot designed to support home healthcare by providing safe, intelligent, and empathetic assistance to users with medical or mobility needs.

The goal of A.L.T.R.U.S is to create a robust robot that can:

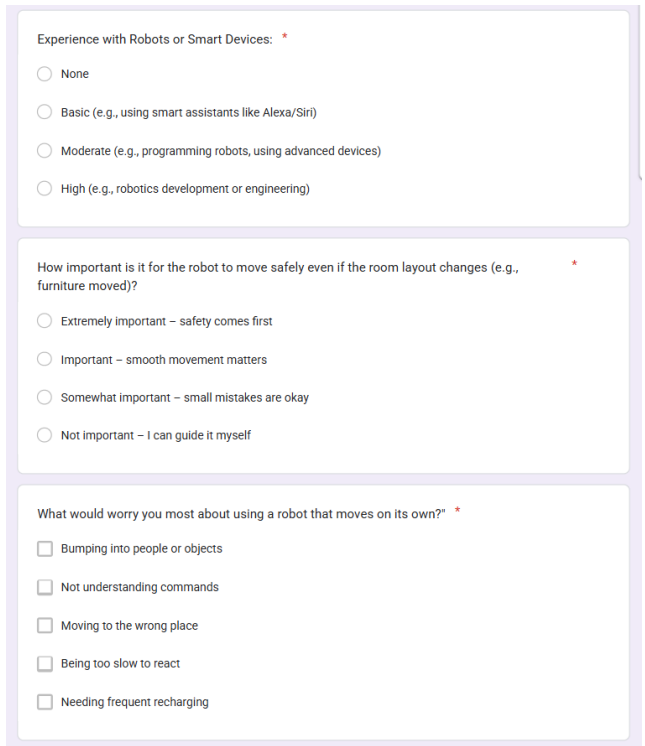
- Understand and respond to voice and gesture commands naturally
- Monitor health and environmental conditions in real time
- Recognize user emotions and respond empathetically
- Recover from faults and update itself securely

The Four Core Components of A.L.T.R.U.S

Voice-Based Command Interpretation & Adaptive Navigation : The robot understands voice commands and uses smart navigation to move safely around obstacles and adapt to changes in the home environment.
Health and Environmental Monitoring Using Sensor Fusion : Multiple sensors work together to continuously monitor vital signs (like heart rate and temperature) and environmental factors (like gas leaks or air quality) for early emergency detection.
Emotion Detection & Empathetic Interaction : The robot recognizes emotions from facial expressions and voice tone, combining these inputs to respond with empathetic speech and appropriate gestures.
Fault-Tolerant Middleware, Secure Updates & Blockchain Logging : A fault-tolerant system ensures continuous operation even during errors, securely manages software updates, and maintains tamper-proof logs of the robot's activities.

Your feedback is crucial to help us understand which features matter most to potential users. The survey will take about 10 minutes to complete.

Thank you for contributing to this important research!



Experience with Robots or Smart Devices: *

- None
- Basic (e.g., using smart assistants like Alexa/Siri)
- Moderate (e.g., programming robots, using advanced devices)
- High (e.g., robotics development or engineering)

How important is it for the robot to move safely even if the room layout changes (e.g., furniture moved)? *

- Extremely important – safety comes first
- Important – smooth movement matters
- Somewhat important – small mistakes are okay
- Not important – I can guide it myself

What would worry you most about using a robot that moves on its own?" *

- [ ] Bumping into people or objects
- [ ] Not understanding commands
- [ ] Moving to the wrong place
- [ ] Being too slow to react
- [ ] Needing frequent recharging

If the robot could understand both your voice and gestures, would that make it easier for you to use? *

- Yes, much easier
- Yes, a little easier
- No difference
- No, I prefer only one control method

Which features would you like most in a home robot with voice and gesture control? *

- [ ] Understand my voice commands accurately
- [ ] Recognize my hand gestures quickly
- [ ] Avoid obstacles without my help
- [ ] Adapt if the furniture is moved
- [ ] Come to me when I call it

What concerns would you have about using a multi-emergency detection wristband?* *

- [ ] Accuracy of health data
- [ ] Privacy of personal health information
- [ ] Battery life
- [ ] Comfort and wearability
- [ ] Cost of the device
- [ ] Risk of false alarms

Have you ever interacted with a robot or virtual assistant? *

- Yes
- No

How would you rate your comfort level interacting with robots? *

- Very comfortable
- Somewhat comfortable
- Neutral
- Uncomfortable
- Very uncomfortable

Which emotions do you think are most important for a robot to detect? *

- [ ] Happiness
- [ ] Sadness
- [ ] Anger
- [ ] Fear
- [ ] Surprise
- [ ] Disgust

In your opinion, which medical emergencies should a wearable health monitoring wristband prioritize detecting? *

Short-answer text

In your opinion, which medical emergencies should a wearable health monitoring wristband prioritize detecting? *

- Very familiar, I use one regularly
- Somewhat familiar, I've heard about them but never used one
- Not familiar at all

If a wristband alerted you about a possible medical emergency, how likely are you to act immediately (e.g., call emergency services, inform family)? *

- Very likely
- Likely
- Neutral
- Unlikely
- Very unlikely

Which of these features would you find most useful in a health emergency detection wristband? *

- [ ] Heart attack detection
- [ ] Tachycardia & bradycardia detection
- [ ] Cardiac arrest detection
- [ ] Fall detection
- [ ] Fever detection
- [ ] Emergency call/SMS alerts
- [ ] Historical health trend analysis

Which emotional cues do you think a robot should detect? *

- [ ] Facial expression
- [ ] Tone of voice
- [ ] Gestures/body language
- [ ] Words used in speech

Would you trust a robot that can read your emotions? *

- ( ) Yes
- ( ) Maybe
- ( ) No

If a robot was helping you at home, what tasks would you want it to do without ever failing? *

Short-answer text

How important is it for an assistive robot to keep working even if some parts stop working? *

- ( ) Extremely important – safety comes first
- ( ) Important – reliability matters
- ( ) Somewhat important – minor issues are okay
- ( ) Not important – I can manage without it

If a robot kept a secure, unchangeable record of what it did, would this make you trust it more? *

- ( ) Yes, a lot more
- ( ) Yes, a little more
- ( ) No difference
- ( ) No, it wouldn't matter

Which of these would you want in a helpful robot? *

- [ ] Understand my requests correctly
- [ ] Fix itself when something goes wrong
- [ ] Keep a secure record of actions
- [ ] Connect different robot features smoothly (e.g., movement + health checks)
- [ ] Share information instantly between parts

What would worry you about using a robot that makes its own decisions? *

- [ ] Safety during malfunctions
- [ ] Misunderstanding commands
- [ ] Privacy of personal information
- [ ] Relying too much on technology
- [ ] Cost of buying or maintaining it
- [ ] Being too complicated to use

From a robotics systems perspective, what core services should middleware guarantee in a mission-critical assistive robot? *

Short-answer text

In modular assistive robots, where would you rank fault tolerance in the middleware priority list? *

- ( ) Top priority – essential for safe autonomy
- ( ) High priority – crucial for user trust
- ( ) Medium priority – nice to have but not critical
- ( ) Low priority – can be handled at application level

How effective do you think blockchain is for ensuring data integrity and accountability in distributed robotic systems? *

- ( ) Very effective – ideal for immutable logging
- ( ) Effective – works well with limitations
- ( ) Neutral – other methods are equally good
- ( ) Ineffective – adds unnecessary complexity

Which features would you expect from a next-generation robot middleware? *

- [ ] Intent recognition and context mapping
- [ ] Real-time fault detection and self-healing
- [ ] Blockchain-based action/event logging
- [ ] Modular plugin management for robot capabilities
- [ ] Developer tools for monitoring and debugging

What challenges do you foresee in implementing fault-tolerant, blockchain-backed middleware for modular robots? *

- [ ] Real-time performance overhead
- [ ] High resource consumption (CPU, storage)
- [ ] Complexity of fault recovery algorithms
- [ ] Blockchain scalability and latency
- [ ] Interoperability between heterogeneous modules
- [ ] Developer adoption and learning curve